

Far Eastern University

THESIS TITLE

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An undergraduate thesis submitted to the

Department of Mathematics

Institute of Arts and Sciences

Far Eastern University

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Bachelor of Science in Applied Mathematics

(Information Technology Track)

May 2022

Declaration of Authorship

We, Juan L. Dela Cruz, Jr., Juan L. Dela Cruz, Jr., and Juan L. Dela Cruz, Jr., declare that this thesis titled, Thesis Title, and the work present in it are our own. We confirm that:

- This work was done wholly or mainly while in candidature for an undergraduate degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where we have consulted the published work of others, this is always clearly attributed.
- Where we have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely our work.
- We have acknowledged all main sources of help.
- Where the thesis is based on work done by ourselves jointly with others, we have made clear exactly what was done by others and what we have contributed ourselves.

Signed: _____

Date: _____

Endorsement

This is to certify that this undergraduate thesis titled, Thesis Title, prepared and submitted by Juan L. Dela Cruz, Jr., Juan L. Dela Cruz, Jr., and Juan L. Dela Cruz, Jr., in partial fulfillment of the requirements for the degree of Bachelor of Science in Applied Mathematics (Information Technology Track) is hereby accepted.

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The Department of Mathematics endorses the acceptance of this undergraduate thesis in fulfillment of the requirements for the degree of Bachelor of Science in Applied Mathematics (Information Technology Track).

Genesis John G. Borja
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APPROVAL SHEET

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Far Eastern University
Institute of Arts and Sciences
Department of Mathematics

Bachelor of Science in Applied Mathematics
(Information Technology Track)

Abstract

Thesis Title

by
Juan L. Dela Cruz, Jr.
Juan L. Dela Cruz, Jr.
Juan L. Dela Cruz, Jr.

Type your abstract here. Your abstract should have a minimum of 150 words and a maximum of 300 words.

Acknowledgments

Type your acknowledgements here. It is customary to acknowledge special assistance from extramural agencies. There is no obligation that assistance received from members of the dissertation or thesis committee be acknowledged.
Acknowledgments should be couched in terms consistent with the scholarly nature of the work. Your name and the date should not appear on this page.

Contents

Abstract v

Acknowledgments vi

List of Tables viii

List of Figures ix

Chapter 1. How to Wrtie Text and Type Equations 1

Chapter 2. How to Compile and Print 3

Chapter 3. How to include JPEGs and Tables 4

Chapter 4. How to Cite References 6

Appendix A.ThesisMain.tex File 9

A.1 Preamble 9

A.2 Basic Information 10

A.3 Abstract and Acknowledgements 12

A.4 Chapters and Appendix 12

A.5 Document Outline 12

A.6 Bibliography 14

A.7 Folder Contents 14

Appendix B.feumathstyle.sty file 16

Bibliography 17

List of Tables

3.1 This is how you caption tables 5

List of Figures

3.1 Secant lines forming the tangent line. 4

Chapter 1

How to Wrtie Text and Type Equations

Write text the way you usually write your documents. Separate paragraphs using the command. Leave an extra space before you write the next paragraph.

To write an equation like $x^2 - 2x + 6 - 8$ within the text, use

```
$x^2 - 2x +6 - 8$
```

In some cases, like writing equations with the summation, integral or limit symbols, include in the equation the \ displaystyle command. For example,

```
This is what happens:
$\displaystyle \sum_{n=0}^\infty \frac{x^n}{n!}$.
```

gives the following result:

This is what happens: $\displaystyle \sum_{n=0}^\infty \frac{x^n}{n!}$.

This is how you write an equation on a different line.

$$f_x + f_{xyy} + \sum_{k=0}^\infty \sqrt{4} + \frac{5}{6}$$

$$\sum_{n=0}^{+\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

```
$$ {f_x} + {f_{xyy}} + \sum_{k=0}^\infty \sqrt{4} + \frac{5}{6} $$
$$ \sum_{n=0}^{+\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n $$
```

$$\begin{aligned} f(x) &= \sum_{n=0}^{+\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \\ &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots \end{aligned}$$

```
\begin{eqnarray*}
f(x)  &=&  \sum_{n=0}^{+\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n\\
&=& f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2+ \ldots
\end{eqnarray*}
```

Separate paragraphs by using `\medskip`, then one more vertical space. You may also use bigskip or smallskip.

This is how you number an equation:

```
\begin{equation}
\int_{-\infty}^\infty e^{x^2/2} dx \label{myequation}
\end{equation}
```

$$\int_{-\infty}^{\infty} e^{x^2/2} dx \tag{1.1}$$

Now, we can reference the equation above using the code: `(\ref{myequation})`. Using this we get equation (1.1).

To work on a theorem environment, you type:

```
\begin{theorem}\label{thm:main}
[Main Theorem] The following is a theorem.
\end{theorem}
```

to get something like this:

Theorem 1.1 *[Main Theorem] The following is a theorem.*

You may replace 'theorem' by 'definition', 'lemma' or 'proposition', or even 'proof':

Chapter 2

How to Compile and Print

If you are using TeXnicCenter, the documents are all linked to a project. This makes the compiling easier. Open the files using "Open project". To compile use F7 or Build output. This will compile to a readable format, depending on which output file you're using (DVI,PS,PDF).

If you are in command line mode or using linux, use the command `latex main`. This command produces the file called `main.dvi`, and to view the dvi file, use the command `xdvi main`.

Print your file using the output file you are using.

Chapter 3

How to include JPEGs and Tables

In this Chapter, you will learn how to include figures in JPEG. In the example below, the jpeg file has to be in the same directory as your tex files. The code is:

```
\begin{figure}[H]
\begin{center}
\includegraphics[width=6cm,height=4cm]{deriv1.jpg}
\end{center}
\caption{Secant lines forming the tangent line.}
\label{fig:deriv1}
\end{figure}
```

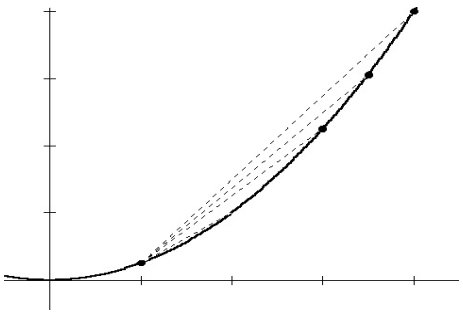


Figure 3.1: Secant lines forming the tangent line.

The figure above is referenced using the `ref` command: Figure 3.1.

It is recommended that you prefix all figure labels by `fig:`, all chapter labels by `sec:`, and all table labels by `tab:`.

For tables, the codes are as follows:

```
\begin{table}[H]
\begin{center}
\begin{tabular}{|l|ll|}
\hline name &  age& sex \\
\hline John & 32 & male \\
       Diwata & 18 & male \\ \hline
\end{tabular}
\end{center}
\caption{This is how you caption tables.}\label{tab:table1}
\end{table}
```

Table 3.1 below illustrates the created table.

name	age	sex
John	32	male
Diwata	18	male

Table 3.1: This is how you caption tables

Chapter 4

How to Cite References

First edit the file `bibliog.bib` by adding an entry for each reference used in your thesis. Follow the format in the sample file.

```
@BOOK{bsw,
  author={H. T. Banks and R. C. Smith and Y. Wang},
  title={Smart Material Structures: Modeling Estimation
         and Control},
  year=1996,
  publisher={Masson/John Wiley},
  address={Paris/Chichester}
}

@ARTICLE{BanksIto1,
  author={H. T. Banks and K. Ito},
  year=1997,
  title={Approximation in {LQR} Problems for Infinite Dimensional
    Systems with Unbounded Input Operators},
  journal={Journal of Mathematical Systems, Estimation, and Control},
  volume=7,
  pages={1-34},
  publisher={Birkh\"{a}user Boston}
}

@PHDTHESIS{Ric,
```

```
  author = {R. C. H. del Rosario },
  year = 1998,
  title = {Computational Methods for Feedback Control in
        Structural Systems},
  school = {N. C. State University}
}

@techreport{RalphRic3a,
  author = {R. C. H. del Rosario and R. C. Smith},
  year = 1997,
  title = {{LQR} Control of Thin Shell Dynamics: Formulation and
           Numerical Implementation},
  institution = {Institute for Computer Applications in Science
    and Engineering},
  number={ICASE Report 97-59},
  address={NASA Langley Research Center, Hampton},
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%    %MISC    %%%%%%%%%%%%%%%%%%%%%%%%%%%%

@BOOKLET{Le,
  author={A. W. Leissa},
  title={Vibrations of Shells},
  year=1973,
  publisher={NASA SP-288},
  howpublished={Reprinted by the Acoustical Society of America
   through the American Institute of Physics}
}
```

```
@manual{d31ref,
  organization = {{Institute} of Electrical and
   Electronics Engineers},
  year = 1978,
  title = {IEEE Standard on Piezoelectricity},
  address={New York}
}

@manual{pside,
  title = {Specification of PSIDE},
  author={Jacques J.B. de Swart and Walter M. Lioen and
   Wolter A. van der Veen},
  year=1998,
  note={Avalable at http://www.cwi.nl/cwi/projects/PSIDE/}
}
```

Click from the BibTex icon (if you are using WinEdit Program) or use the BibTex command if you are in command line mode. It is suggested that you do this twice. Then compile the main tex file using the LaTeX command.

Use the `cite` command to cite a reference. For example, `\cite{bsw}` cites the book labeled "bsw". This will give us (Banks, Smith, & Wang, 1996).

Appendix A

ThesisMain.tex File

The ThesisMain.tex file contains the basic information about your thesis while the chapter files are just called in the code. This is the only file you need to compile. Do not run the chapter files.

A.1 Preamble

```
\documentclass[12pt]{report}
\usepackage{feumathstyle}
\usepackage{amssymb, graphicx, amsmath}
\usepackage{hyperref}
\usepackage{apacite}
\bibliographystyle{apacite}

\usepackage{lscape}

\newtheorem{definition}{Definition}[chapter]
\newtheorem{theorem}{Theorem}[chapter]
\newtheorem{lemma}[theorem]{Lemma}
\newtheorem{corollary}[theorem]{Corollary}
\newtheorem{proposition}[theorem]{Proposition}
```

you may add additional packages but do not remove the prelisted packages especially the feumathstyle package. You may add other new formats here too

A.2 Basic Information

```
\title{Thesis Title} %do not capitalize all

\authorfirst{Juan L.}
\authorlast{Dela Cruz}
\authorext{Jr.}
\authorexttrue %make this a comment if you do not have an extension name e.g. Jr.,

\secondauthorfirst{Juan L.}
\secondauthorlast{Dela Cruz}
\secondauthorext{Jr.}
\secondauthorexttrue %make this a comment if you do not have an extension name e.g

\thirdauthorfirst{Juan L.}
\thirdauthorlast{Dela Cruz}
\thirdauthorext{Jr.}
\thirdauthorexttrue %make this a comment if you do not have an extension name e.g.

\degree{Bachelor of Science in Applied Mathematics } %enter the specific degree
\thesistype{thesis}
\thesisdocument{undergraduate thesis}

\specialization{with Information Technology} \specializetrue

\defensedate{May 12, 2022} %indicate the month, day and year the research was defe
\submitmonth{May} %indicate the month the manuscript was submitted
\submityear{2022} %indicate the year the manuscript was submitted
```

```
\adviser{Adviser, Ph.D.}%main adviser
\panel{Panel, Ph.D.}
\secondpanel{Panel, Ph.D.}
\thirdpanel{Panel, Ph.D.}


\thesisinstructor{May Anne C. Tirado, Ph.D.}


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% highlight \bothabsenttrue if the student has only one adviser and one reader
%% otherwise comment \bothabsenttrue and highlight \coadviser or \coreader which
%% ever is applicable. In this case, the package assumes either you have 1 Adviser
%% and 2 Readers, or 2 Advisers and 1 Reader.

\bothabsenttrue

%\coadviser{co-Adviser, Ph.D.}
%\coadvisetrue
% Make the \coadvisetrue as a comment if there is no adviser
%%%%%%%%%%%%%%%%%%%%%%%%%%%%


\deptchair{Genesis John G. Borja}
\dean{Dean Name}
```

input the names and information about your thesis, authors, adviser, panel, dean.

A.3 Abstract and Acknowledgements

```
\abs{Type your abstract here.  Your abstract should have a minimum
of 150 words and a maximum of 300 words.}

\acknowledge{Type your acknowledgements here. It is customary to acknowledge specia
Acknowledgments should be couched in terms consistent with the
scholarly nature of the work. Your name and the date should not appear on this pag
```

type your abstract and acknowledgements here

A.4 Chapters and Appendix

Type your chapters and appendix in separate tex files. This is so it will be easier to edit a shorter code than the full manuscript.

The chapter files starts with

```
\chapter{How to Wrtie Text and Type Equations}
\label{chap:typing}
```

The label is optional. This part is when you want to cite this chapter in other chapters. This file just ends with the last line of your chapter content.

A.5 Document Outline

```
\beforepreface
\abstractpage

\acknowledgetrue
\acknowledgepage
```

```
\tableofcontents

\newpage


\listoftables
\listoffigures

\afterpreface

\include{1typing}
\include{2compiling}
\include{3figures}
\include{4citing}
%%%%% edit the chapter files. no more preamble. start directly on \chapter{}. add r

\appendix
\input{appendixA}
\input{appendixB}
%%%%% edit appendix files. no more preamble. start directly on \chapter. you may re

\nocite*
\bibliography{BIBLIOG.bib}
%to edit your bibliography, open the BIBLIOG.BIB file.
```

The first set of commands will create beforepreface using the information you inputed at the start of ThesisMain.tex. Include the chapters in order and input the appendices in order too. The chapter file names are used in the arguments. It is best to not use

space in the chapter file names.

A.6 Bibliography

Input your references in the Bibliography.bib file

```
@article{Datta-Figueira-2013,
author = {Dilip Datta and Jos\'e Rui Figueira},
title = {{A real-integer-discrete-coded differential evolution}},
journal = {Applied Soft Computing},
volume = {13},
number = {9},
pages = {3884--3893},
year = {2013}
}
@book{Deb-2001,
author = {Kalyanmoy Deb},
title = {{Multi-Objective Optimization using Evolutionary Algorithms}},
publisher = {John Wiley \& Sons Ltd.},
address = {Chichester, England},
year = {2001}
}
```

A.7 Folder Contents

All your thesis files must be in one folder before you compile/run ThesisMain.tex

1. ThesisMain.tex
2. feumathstyle.sty

3. FEUlogo.png
4. IASlogo.png
5. BIBLIOG.bib
6. chapter tex files
7. appendix tex files

See the sample compiled ThesisMain.pdf to see the intended output.

Appendix B

feumathstyle.sty file

DO NOT EDIT feumathstyle.sty unless necessary. If you want to edit beforepreface contents that are not on the main tex file, and other formats, do it on feumathstyle.sty file. Be careful in editing this file. You may create a copy first and edit the copy.

feumathstyle.sty contains the text format and layout of the beforepreface pages, chapter pages, appendices, table of contents and other lists.

the definition of some commands are also on this file.

Bibliography

Banks, H. T., Smith, R. C., & Wang, Y. (1996). *Smart material structures: Modeling estimation and control*. Paris/Chichester: Masson/John Wiley.

Burke, E., Elliman, D., & Weare, P. F. R. (1996). Examination Timetabling in British Universities - A Survey. In E. K. Burke & P. Ross (Eds.), *Proceedings of practice and theory of automated timetabling* (Vol. 1153, pp. 76–90). Springer.

Colorni, A., Dorigo, M., & Maniezzo, V. (1992). *A Genetic Algorithm to Solve the Timetable Problem* (Tech. Rep. No. 90-060 revised). Politecnico di Milano, Italy.

Datta, D. (1998). *Optimal Shape Design System for Plates under Dynamic Loads* (Unpublished master's thesis). Indian Institute of Technology, Delhi. (Master thesis)

Datta, D., & Figueira, J. R. (2013). A real-integer-discrete-coded differential evolution. *Applied Soft Computing*, *13*(9), 3884–3893.

Deb, K. (2001). *Multi-Objective Optimization using Evolutionary Algorithms*. Chichester, England: John Wiley & Sons Ltd.